Andrea Wan

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PROFESSIONAL SUMMARY

- **Utility Infrastructure Engineering:** Hands-on experience supporting utility infrastructure projects, including substation transformer installations, AC power modifications, lighting upgrades, and field implementation activities through drawing review, wiring verification, and technical coordination.
- **Mechanical & Eletrical Design Support:** Skilled in engineering calculations, electrical schematic review, equipment analysis, load verification, and mechanical system improvement projects using engineering drawings and technical documentation.
- **Project Execution & Coordination:** Experienced in supporting multidisciplinary engineering projects from design development to field execution, collaborating with engineers, designers, and stakeholders to deliver reliable infrastructure solutions.
- **Asset Management & Engineering Automation:** Proficient in SAP, Microsoft Excel, VBA, and Power BI for equipment data management, dashboard automation, reporting optimization, and improving engineering workflow efficiency.

PROFESSIONAL EXPERIENCE

Engineering Intern - Substation **May 2026 – Present**
Fortis BC **Trail, BC**

- **Field Verification:** Conduct field wire checks for 5+ substation transformer installations, verifying wiring, terminations, and connections against engineering drawings and design requirements.
- **Engineering Calculations:** Support 15+ operating limit and engineering calculation packages, compiling equipment data and assisting with technical verification and documentation.
- **Material Management:** Develop and standardize 5+ substation material lists using Microsoft Excel and SAP, improving consistency and supporting material tracking and procurement activities.
- **Drawing Review:** Review electrical and substation engineering drawings, identifying wiring, equipment, and connection details requiring clarification or revision.
- **Project Execution:** Support multiple substation engineering projects from design through field implementation, coordinating engineering requirements, material information, drawing updates, and field activities.
- **Technical Documentation:** Prepare and maintain 30+ technical documents and engineering records using SAP and Microsoft Office, supporting accurate project documentation and engineering deliverables.
- **Design Coordination:** Collaborate with engineers, designers, and drafters to support design updates, drawing revisions, material coordination, and resolution of field engineering issues.

Engineering Intern **March 2024 – June 2024**
China Penghui Environmental Protection Engineering Ltd. **China**

- **Operational Infrastructure Maintenance:** Optimized the performance of water quality monitoring stations, ensuring 95% uptime for critical environmental data collection sensors.
- **Wastewater Treatment Projects:** Supported large-scale treatment projects by recording operational data for 10+ facilities, leading to a 12% improvement in reporting accuracy.
- **Equipment Reliability Assurance:** Conducted rigorous inspections of mechanical components, identifying potential failure points and reducing emergency repair costs by $15,000 per quarter.
- **Technical Documentation Management:** Standardized documentation protocols for site visits and equipment logs, streamlining the audit process for senior engineers by 25%.
- **Cross-Functional Team Collaboration:** Assisted lead project managers in the execution of 3 major municipal wastewater upgrades, meeting all technical milestones 1 week ahead of schedule.
- **Environmental Standard Compliance:** Monitored station output against national environmental regulations, ensuring 100% compliance with government water safety thresholds.

PROJECTS

Station Lighting Upgrade Packages | Fortis BC **2026**

- **Strategic Project Planning:** Managed 3 comprehensive lighting upgrade packages for Pine Street, Lambert, and SLC2 stations, ensuring 100% adherence to rigorous delivery schedules.
- **Efficiency Delivery Approach:** Implemented a simplified work summary package methodology, reducing administrative overhead by 25% compared to traditional IFC packages.
- **As Built Integration:** Analyzed historical As-Built documentation to inform modern lighting designs, achieving a 100% success rate in station illumination compliance.

Sound-Responsive Automatic Page Turner | UBC **2026**

- **Mechatronic Design Integration:** Engineered an ESP32-based automated system for musicians, utilizing MIDI and audio sensing to achieve a 98% successful page-turn rate.
- **Digital Signal Processing:** Implemented real-time moving average filters and frequency feature extraction to reduce ambient noise interference by 40% during performances.
- **Hardware Prototyping:** Designed and 3D-printed mechanical components in SolidWorks, optimizing motor torque and reducing device weight by 15%.

Substation Asset Management Dashboard Automation | Fortis BC **2026**

- **Data Integration & Automation:** Automated SAP equipment data extraction and Excel-based data processing workflows to improve substation asset information management and reduce manual data handling.
- **Dashboard Development & Visualization:** Developed interactive Excel dashboards using formulas, pivot tables, and data visualization tools to track equipment status, material information, and engineering activities.
- **Reporting & Workflow Optimization:** Streamlined engineering reporting processes by improving data accuracy, standardizing asset records, and enhancing visibility for project coordination and planning.

Line-Following Autonomous Robot | UBC **2025**

- **Autonomous Navigation:** Developed a high-precision Arduino and IR sensor line-following algorithm, achieving 90% path-tracking accuracy at peak speeds while maintaining stable navigation through complex track layouts.
- **Differential Motor Control:** Programmed differential motor control sequences to improve cornering stability by 25%, enabling reliable navigation through sharp 90-degree turns and varying track conditions.
- **Embedded Mechanical Integration:** Designed a SolidWorks 3D-printed obstacle-detection mechanism and custom PCB layouts for high-current DC motors, achieving 100% detection reliability while reducing electrical interference by 30%.

Creative Custom Feline Infrastructure | UBC **2024**

- **Ergonomic Structural Modeling:** Designed dual custom cat trees using CAD to optimize load distribution for 25lb weights, ensuring a 2.5x safety factor for structural stability.
- **High-Precision Fabrication:** Executed precision woodworking to ±1 mm tolerances, integrating reinforced framing that extended the product lifespan by 3x over retail alternatives.
- **Path Optimization Simulation:** Leveraged 3D modeling to simulate feline climbing paths, resulting in a 20% increase in usable space through efficient vertical design.

EDUCATION

Bachelor of Applied Science in Integrated Engineering **Sep 2024 - May 2029 (Expected)**
University of British Columbia | CGPA: 82.6% (A-) Vancouver, BC

Bachelor of Science in Mathematics, Operational Research, Statistics, and Economics **2022 – 2023**
University of Warwick Coventry, UK

TECHNICAL SKILLS

Mechanical Systems Design: SolidWorks (CSWA Certified), 3D Modeling & 2D Drawing, Tolerance Stack-Up Analysis, Material Property Analysis, Basic Machining
Electrical & Electronics: Microcontroller Programming (Arduino Uno, ESP32), Circuit Soldering & Prototyping, Substation Design, Power Distribution Calculations, Electrical Schematic Design
Programming & Software: C++, C#, Java, Microsoft Excel (Advanced Data Modeling), VBA, Power BI Bluebeam Revu, SAP, JDE
Project Management: Project Planning & Scheduling, Technical Documentation, Cross-Functional Coordination, Risk Assessment, Progress Tracking, Engineering Change Management, Stakeholder Communication